

# Machine learning

Decision tree regression

# Load dataset

# Importing the libraries

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import pandas as pd
```

# Importing the dataset

```
dataset = pd.read_csv('Position_Salaries.csv')
```

# Dependent & independent variables

```
X = dataset.iloc[:, 1:2].values
```

```
y = dataset.iloc[:, 2].values
```

# Training and testing sets

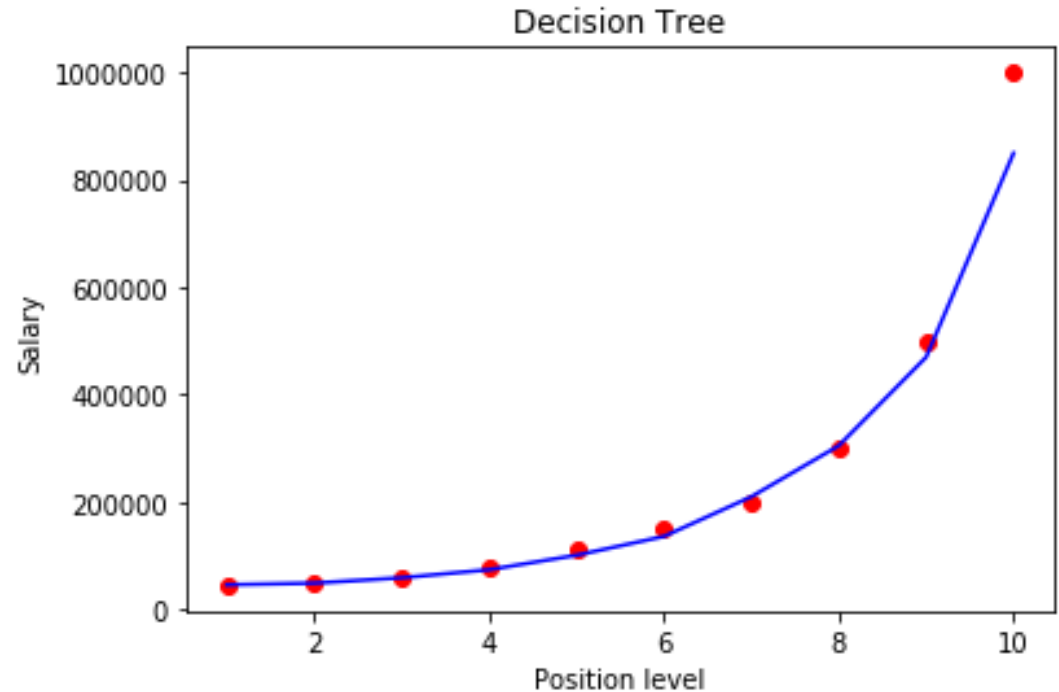
Not enough Data....

# Fitting the decision tree to the data

```
from sklearn.tree import DecisionTreeRegressor  
  
regressor = DecisionTreeRegressor(random_state=0)  
  
regressor.fit(X, y)
```

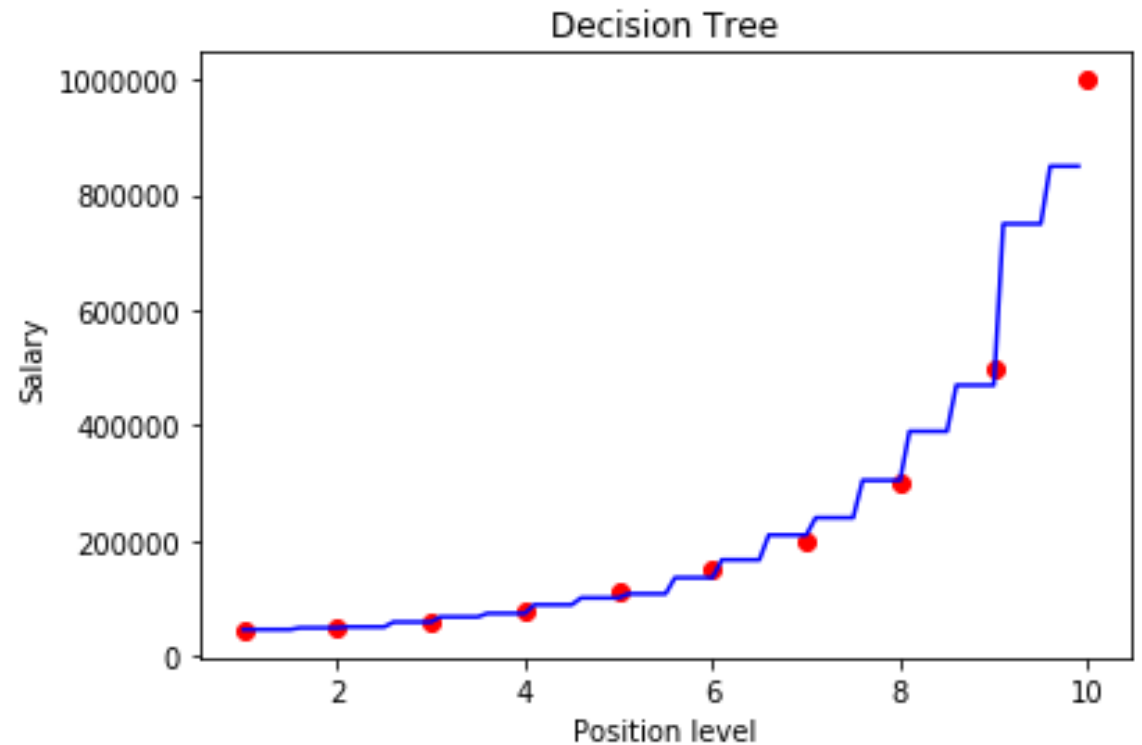
# Visualizing the results - trap!

```
plt.scatter(X, y, color = 'red')  
plt.plot(X, regressor.predict(X),  
color = 'blue')  
plt.title('Decision Tree')  
plt.xlabel('Position level')  
plt.ylabel('Salary')  
plt.show()
```



# Visualizing the results

```
X_grid = np.arange(min(X), max(X), 0.1)  
    #we get a vector  
X_grid = X_grid.reshape((len(X_grid), -1))  
    #we need a matrix  
plt.scatter(X, y, color = 'red')  
plt.plot(X_grid, regressor.predict(X_grid),  
color = 'blue')  
plt.title('Decision Tree')  
plt.xlabel('Position level')  
plt.ylabel('Salary')  
plt.show()
```



```
text_representation =  
tree.export_text(regressor)
```

```
|--- feature_0 <= 8.50  
| |--- feature_0 <= 6.50  
| | |--- feature_0 <= 4.50  
| | | |--- feature_0 <= 3.50  
| | | | |--- feature_0 <= 2.50  
| | | | | |--- feature_0 <= 1.50  
| | | | | | |--- value: [45000.00]  
| | | | | | |--- feature_0 > 1.50  
| | | | | | |--- value: [50000.00]  
| | | | |--- feature_0 > 2.50  
| | | | |--- value: [60000.00]  
| | | |--- feature_0 > 3.50  
| | | |--- value: [80000.00]  
| | |--- feature_0 > 4.50  
| | |--- feature_0 <= 5.50  
| | | |--- value: [110000.00]  
| | | |--- feature_0 > 5.50  
| | | |--- value: [150000.00]  
| |--- feature_0 > 6.50  
| | |--- feature_0 <= 7.50  
| | | |--- value: [200000.00]  
| | | |--- feature_0 > 7.50  
| | | |--- value: [300000.00]  
|--- feature_0 > 8.50  
| |--- feature_0 <= 9.50  
| | |--- value: [500000.00]  
| |--- feature_0 > 9.50  
| | |--- value: [1000000.00]
```



# תרגיל כיתה

► הריצו עץ רגרסיה על הקובץ `Position_Salaries.csv`

► כאשר אתם מייצרים את הרגרסור ישנם עוד פרמטרים שניתן לשחק איתם (למשל: `max_depth`).  
קראו את ההסברים על הפרמטרים ונסו באמצעותם להבין כיצד מתבצעת החלוקה של העץ.

# Random forest

The same as before with a little change:

```
from sklearn.ensemble import RandomForestRegressor  
regressor=RandomForestRegressor(n_estimators=10, random_state=0)  
regressor.fit(X,y)  
  
pred=regressor.predict([[6.5]])
```

# תרגיל כיתה

- ▶ הריצו random forest עם 3 עצים על הקובץ Position\_Salaries.csv
- ▶ שחקו עם עומק היער וצרו פרדיקציות שונות